higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PLANT AND EQUIPMENT
NQF LEVEL 2

20 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1: INTRODUCTION TO PLANT AND MACHINERY

1.1 True
1.2 True
1.3 False
1.4 True
1.5 True

(5 × 1) **[5]**

QUESTION 2: LEGISLATION AND SAFETY REGULATIONS

2.1
- How construction workers must behave on construction sites
- How they must use their tools and equipment
- How they must handle their building materials
- The type of personal protective equipment (PPE) they must wear and use
- The quality of their building materials and work

(Any relevant answers accepted) (5)

2.2 To protect construction workers✓ and to allow them to receive compensation✓ if they are injured in an accident on a construction site.✓

(Any relevant answers accepted) (3)

2.3 2.3.1 The Act excludes workers in mines✓ and on ships✓ – other laws apply to these workers. (2)

2.3.2
- Farm workers
- Domestic workers
- State workers.
- Construction workers (Any 3 × 1) (3)

2.4 2.4.1 Personal protective equipment
2..2 Safe erection method statement

(2 × 1) (2)

[15]

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QUESTION 3: FIRST AID

3.1
- Highest priority✓ – people who need immediate help and transport to medical help✓
- Second priority✓ – people who can probably wait for up to an hour for medical help without it posing a risk to their lives✓
- Lowest priority✓ – people who can wait and receive transport last or those who are obviously dead✓ (3 × 2) (6)

3.2
- On the wrist
- On the neck (2)

3.3
- Colour of the skin (pale or grey)
- Presence of perspiration (Any relevant answers accepted) (2)

3.4
- First-degree burn
- Second-degree burn
- Third-degree burn (3)

3.5 Completely clean✓ and free of bacteria✓ (2)

[15]

QUESTION 4: HAND TOOLS

4.1
- Cross-cut saw
- Rip saw
- Tenon saw/back saw
- Coping saw
- Pad saw
- Compass saw
- Hand saw
- Jig saw (Any 5 × 1) (5)

4.2 It will be used when you have dropped a small metal object or a metal tool into a part of an engine that you cannot reach with your hand.
(Any relevant answers accepted) (2)

4.3
- Pentagon
- Hexagon
- Octagon (3)

4.4
- First, consult✓ the building or the floor plan.✓
- Secondly, get the formwork and support work specifications✓ from the engineer.✓
- Then make sure that you have all the materials, equipment and tools✓ that you will need to mark, set out and prepare the shuttering that will make up your formwork.✓
- A traditional formwork technique must be used,✓ which consists of erecting form faces from the outside using pegs.✓
- Because wet concrete is very heavy,✓ your formwork must be able to withstand the pressure that the wet concrete puts on it.✓

(Any relevant answers – 5 × 2) (10)

4.5 Take two pieces of timber that are longer and thicker than the oilstone to form the base and lid of the box.✓ Centre the oilstone on the base piece and mark it out.✓ Use the marking gauge to mark out the size of the stone, which will help guide the chisel.✓ In the same direction as the grain, slowly and gently chisel out the recess until the recess is about half the height of the stone.✓ Place this base alongside the timber that will become the lid and use a square to transfer the same measurements across from the base.✓ Use the gauge again to mark out the chisel lines.✓ Chisel out about half the height of the stone, as done with the base.✓ The stone will now fit snugly between the lid and the base.✓ Using the plane,✓ finish off the edges neatly, as this will give the oilstone box an attractive look.✓ (Any relevant answers accepted) (10)

[30]

QUESTION 5: LADDERS

5.1
- Never place a ladder in front of a doorway.
- Never have two people climbing up a ladder or someone standing below the ladder.
- If possible, the ladder must extent 1 metre above its support.
- When you use a two-legged ladder, you must open it fully and lock its braces.
- Never use ladders horizontally or as scaffolding, because they are not designed for this purpose.
- Ladders must have safety feet to give more stability and to prevent slipping.
- Another person must always hold a straight ladder in position, unless it is securely tied at the top.
- Place a straight ladder in such a way that its feet are a quarter of its length away from the base of the object against which it is resting.

(Any relevant answers – 5 × 1) (5)

5.2 The ladder can be moved sideways along the wall, which may cause you to lose your position on the ladder.✓ The ladder can topple over sideways, causing the ladder and you to fall to the ground.✓ You may lose your balance and you may fall off the ladder.✓ In an attempt to regain your balance, you may drop the tool that you are holding and the tool can fall and break or injure somebody.✓ You may lose your balance and accidentally push against the wall or pole against which the ladder is leaning, causing the ladder to fall backwards.✓ (Any relevant answers accepted) (5)

[10]

QUESTION 6: TRESTLES, SCAFFOLDING AND MOBILE TOWERS

6.1

6.1.1	F
6.1.2	I
6.1.3	A
6.1.4	K
6.1.5	B
6.1.6	L
6.1.7	D
6.1.8	E
6.1.9	C
6.1.10	J

(10 × 1) (10)

6.2
- Platforms that are based on a trestle must be fully boarded, with no gaps between the boards.
- Platforms must be adequately supported.
- You must provide a safe access to trestle working platforms.
- If your trestle working platform is higher than 2,5 metres, you must provide guard rails and toe boards on all the open sides.
- There must be overhead protection if workers are exposed to overhead hazards. (Any relevant answers accepted) (5)

6.3
- The ground must be level.
- The ground over which the tower is moved must be able to support the weight of the tower.
- Never move the tower faster than normal walking speed.
- Towers must only be moved by hand or with moving machines.
- No one is allowed on the tower when the tower is being moved.

(Any relevant answers accepted) (5)

6.4
- Guard rail
- Toe board
- Working platform
- Transom
- Ledgers
- Double coupler
- Secured ladder
- Cross bracing
- Swivel coupler
- Putlog clip
- Base plate

(Any relevant parts for independent scaffolding accepted – 5 × 1) (5)
[25]

TOTAL: 100

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